

16 Hypothesis testing for 1 and 2 samples

Subject content	Additional information
Know how to apply knowledge about carrying out hypothesis testing to conduct tests for the:	
16.1 mean of a normal distribution with unknown variance using the t -distribution	Students are expected to use the t -distribution when the sample size is small ($n < 30$) and the population standard deviation, σ , is unknown. t -test – assumption required is that the distribution is normal for test to be valid.
16.2 difference of two means for two independent normal distributions with known variances	

16 Hypothesis testing for 1 and 2 samples (continued)

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16.3 difference of two means for two independent normal distributions with unknown but equal variances	Students should be aware that a test for a difference of two means for two independent normal distributions with unknown variances is valid only if equal variances can be assumed. Candidates will be expected to use a pooled estimate for an unknown common population variance. Candidates will not be expected to test for equality of variances. <i>t</i>-test – assumptions required are that both distributions are normal and have equal variances for test to be valid.
16.4 difference between two binomial proportions	Candidates will be expected to use a pooled estimate for an unknown population proportion.
16.5 interpret results for these tests in context.	